

Lec 12-2, Nov 26, Wed.

- Improper integrals (8.8)
 - convergence tests: direct CT, limit CT
- First-order differential equations (D.E.) (Chap 9)
 - Overview and examples
 - Definitions
 - Visualizing solutions (9.1)
- Solving first-order D.E.
 - Separable equations (7.4)
 - linear equations (9.2)

Theorem (Direct Comparison Test)

Suppose that $a \in \mathbb{R}$, and suppose that f and g are continuous functions on $[a, \infty)$. If there exists $c \in [a, \infty)$ such that $0 \leq f(x) \leq g(x)$ for all $x \in [c, \infty)$, then the following statements hold:

- (i) If $\int_a^\infty g(x) dx$ converges, then $\int_a^\infty f(x) dx$ converges.
- (ii) If $\int_a^\infty f(x) dx$ diverges, then $\int_a^\infty g(x) dx$ diverges.

"integrable on $[a, b]$, $\forall b \geq a$ "
is enough.

The following is another "comparison test".

Type-I improper integral here

Theorem (Limit Comparison Test)
 Suppose that $a \in \mathbb{R}$, and suppose that f and g are positive continuous functions on $[a, \infty)$. If
"integrable on $[a, b]$, $\forall b \geq a$ "
 is enough.

$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = L$

← f and g grow at the same rate when $x \rightarrow \infty$.

for some $L \in \mathbb{R}_+ := (0, \infty)$, then

$\int_a^\infty f(x) dx$ and $\int_a^\infty g(x) dx$

both converge or both diverge.

• Since $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = L > 0$, $\exists N \in \mathbb{R}$ ($N \geq a$), s.t.

$$\left| \frac{f(x)}{g(x)} - L \right| \leq \frac{L}{2}, \quad \forall x > N \Rightarrow 0 < \frac{L}{2} \leq \frac{f(x)}{g(x)} \leq \frac{3L}{2}, \quad \forall x > N.$$

$$\Rightarrow 0 < \frac{L}{2} g(x) \leq f(x) \leq \frac{3L}{2} g(x), \quad \forall x > N \quad (*)$$

• If $\int_a^\infty f(x) dx$ converges, then $\int_a^\infty \frac{L}{2} g(x) dx$ converges by (*)

based on direct comparison test, so

$$\int_a^\infty g(x) dx = \frac{2}{L} \int_a^\infty \frac{L}{2} g(x) dx \quad \text{converges.}$$

• If $\int_a^\infty f(x) dx$ diverges, then $\int_a^\infty \frac{3L}{2} g(x) dx$ diverges by (*)

based on direct comparison test, so

$$\int_a^\infty g(x) dx = \frac{2}{3L} \cdot \int_a^\infty \frac{3L}{2} g(x) dx \quad \text{diverges.}$$



• The limit comparison test may be more convenient sometimes.

e.g. $\int_1^{\infty} \frac{1-e^{-x}}{x} dx$

$$\begin{aligned} \textcircled{1} \quad \lim_{x \rightarrow 1} f(x) &= f(1) = \frac{1-e^{-1}}{1} < \infty \quad (\text{no Type-II}) \\ \textcircled{2} \quad \lim_{x \rightarrow \infty} f(x) &= \lim_{x \rightarrow \infty} \frac{1-e^{-x}}{x} = 0 \quad (\text{nece' to be convergent}) \end{aligned}$$

Sol: draft: $\textcircled{3} \quad 0 < \frac{1-e^{-x}}{x} < \frac{1}{x}$; $\int_1^{\infty} \frac{1}{x} dx$ diverges!
 $\Rightarrow$ DCT cannot determine.

$\textcircled{4} \quad \frac{\frac{1-e^{-x}}{x}}{\frac{1}{x}} \stackrel{\text{LCT}}{=} 1-e^{-x} \rightarrow 1 (x \rightarrow \infty)$



$\therefore \lim_{x \rightarrow \infty} \frac{\frac{1-e^{-x}}{x}}{\frac{1}{x}} = 1 \in (0, \infty)$

and $\int_1^{\infty} \frac{1}{x} dx$ diverges.

$\therefore$ By limit comparison convergence test,

$\int_1^{\infty} \frac{1-e^{-x}}{x} dx$ diverges as well.

□

Rmks. (1) The limit comparison test can be extended. If f and g

are positive and continuous on $[a, \infty)$, and

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = L,$$

then:

• if $L=0$ and $\int_a^{\infty} g(x) dx$ converges, then $\int_a^{\infty} f(x) dx$ converges ;

• if $L=\infty$ and $\int_a^{\infty} g(x) dx$ diverges, then $\int_a^{\infty} f(x) dx$ diverges .

e.g. f, g unbound at b ,

(2) The test also applies to Type-II improper integrals: $\lim_{x \rightarrow b} \frac{f(x)}{g(x)} = L \in (0, \infty)$.

Extended exercises

(a) Show that $\int_a^\infty e^{-kt} dt$ is convergent for all $a \in \mathbb{R}$ and all $k > 0$.

(b) The **Gamma function** is the function $\Gamma: (0, \infty) \rightarrow \mathbb{R}$ defined by

$$\Gamma(x) \triangleq \int_0^\infty t^{x-1} e^{-t} dt = \int_0^\infty \frac{t^{x-1}}{e^t} dt.$$

[This is an extension of the factorial function to the domain $(0, \infty)$:

one can check that $\Gamma(n+1) = n!$ for any $n \in \mathbb{N} \in \{0, 1, 2, \dots\}$.]

(See Chap 8, additional and advanced exercises Q51.)

$$n! \triangleq n \times (n-1) \times (n-2) \times \dots \times 1$$

Prove that $\int_0^\infty t^{x-1} e^{-t} dt$ converges for each $x > 0$ (so that the Gamma function $\Gamma(x)$ is well-defined).

$$\begin{aligned} \text{Sol: (a)} \quad \int_a^\infty e^{-kt} dt &= \left. \frac{1}{-k} e^{-kt} \right|_a^\infty = -\frac{1}{k} (0 - e^{-ka}) \\ &= \frac{1}{k} e^{-ka} < \infty, \quad \forall k > 0, \\ &\quad a \in \mathbb{R}. \end{aligned}$$

$$(b) \text{ Fix } x > 0. \text{ Then } \Gamma(x) = \underbrace{\int_0^1 t^{x-1} e^{-t} dt}_{I_1} + \underbrace{\int_1^\infty t^{x-1} e^{-t} dt}_{I_2}.$$

It suffices to show that both I_1 and I_2 are finite.

• I_1 : if $x \geq 1$, t^{x-1} and e^{-t} are continuous and bounded as a func' of t .

$\Rightarrow t^{x-1}e^{-t}$ is inte' on $[0, 1] \Rightarrow I_1$ is a definite integral

if $x \in (0, 1)$, $-1 < x-1 < 0 \Rightarrow I_1 < \infty$.

, so $\int_0^1 t^{x-1} dt = \frac{1}{x} t^x \Big|_0^1 = \frac{1}{x} - 0 < \infty$ (conv')

$\therefore 0 < t^{x-1} e^{-t} < t^{x-1} \cdot e^0$, $\forall t \in [0, 1]$, by comparison test,

$\therefore$ we have $\int_0^1 t^{x-1} e^{-t} dt$ converges.

• I_2 : Limit-compare with $e^{-\frac{1}{3}t}$ $\rightarrow \int_1^\infty e^{-\frac{1}{3}t} dt = -3e^{-\frac{1}{3}t} \Big|_1^\infty = 3 < \infty$, which is convergent by (a).

Since $\frac{t^{x-1} e^{-t}}{e^{-\frac{1}{3}t}} = \frac{t^{x-1}}{e^{\frac{2}{3}t}} \rightarrow 0 \quad (t \rightarrow \infty)$.

I_2 is convergent by limit comparison test as $\int_1^\infty e^{-\frac{1}{3}t} dt$ converges. $\square$

• First-order Differential Equations (D.E.)

We have equations like $y' = x^2$. This is an example of a differential equation (D.E.). It has a general solution

$$y(x) = \frac{1}{3} x^3 + C.$$

$$(check: y' = x^2).$$

Population Growth (I) (Malthusian growth model, Thomas Malthus, 1798)

One model for the growth of a population is based on the assumption that the population grows at a rate proportional to the size of the population. If $P(t)$ represents the population at time t , then this model is represented by the differential equation

$$\frac{dP}{dt} = kP \quad (\text{later: } P(t) = P_0 \cdot e^{kt})$$

for some constant k .

$$(check: P'(t) = P_0 k e^{kt} = k \cdot P(t))$$

Population Growth (II) (Logistic growth model, P.F. Verhulst, ~1840)

However, the previous model is too ideal, since population is not likely to grow indefinitely due to limited resources and other constraints. A more practical model would assume:

- ▶ initially, the growth of P is approximately proportional to P ;
- ▶ the growth will decrease as P gets closer to a certain carrying capacity M .

The following differential equation is more consistent with the assumptions above:

$$\frac{dP}{dt} = kP \left(1 - \frac{P}{M} \right). \quad \Rightarrow \quad \underline{P' - kP + \frac{k}{M} P^2 = 0.}$$

↓

$$P' = kP - kP^2 \cdot \frac{1}{M}$$

Here, $k > 0$ is a constant depending on the environment.

$M > 0$ is the carrying capacity.

$K \triangleq k \cdot (1 - \frac{P}{M})$ is the proportionality factor (which is not a const).

$K \approx k$ when P is small ($P \ll M$).

$K \searrow$ as $P \nearrow$.

When $P = M$, $K = 0$ and P stops growing.

When $P > M$, $K < 0$ and P would decrease.

Def: Consider y as a function of x . Let y' mean $\frac{dy}{dx}$.

A first-order differential equation (D.E.) is an equation of the form

$$G(x, y, y') = 0, \quad \forall x \in I. \quad (*)$$

A solution to (*) is a function $y = f(x)$ defined on some interval I s.t.

$$G(x, f(x), f'(x)) = 0, \quad \forall x \in I \quad \cdot \quad (P' - kP + k\frac{1}{M}P^2) = 0$$

The general solution to (*) is the collection of all solutions to (*).

A first-order initial value problem (IVP) is a first-order D.E. together with an

$$\begin{cases} G(x, y, y') = 0 \\ y(x_0) = y_0 \end{cases} \quad (x_0, y_0 \text{ constants})$$

A particular solution is a solution that satisfies the IVP.

A common form of a first-order D.E. is $y' = F(x, y)$.

$$(i.e., G(x, y, y') = y' - F(x, y))$$

E.g.: Consider the D.E. $y' = \frac{x^2}{y^2}$.

Then $y = (x^3 + C)^{\frac{1}{3}}$ is a solution for every constant C .

$$\text{Since } y' = \frac{1}{3} (x^3 + C)^{-\frac{2}{3}} \cdot 3x^2 = \frac{x^2}{(x^3 + C)^{2/3}} = \frac{x^2}{y^2}.$$

$$\Rightarrow y(0) = C^{\frac{1}{3}} = 2 \Rightarrow C = 8$$

If we require $y(0) = 2$ (initial condition), then by solving

for C , we see that $y(x) = (x^3 + 8)^{\frac{1}{3}}$ is the particular solution.

[We will see how one can find the general solution above].

• Visualizing solutions

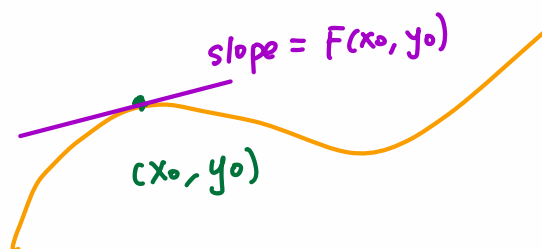
Q: Given a D.E. $\frac{dy}{dx} = F(x, y)$, what do the graphs of its solutions look like?

A: Slope fields.

Suppose $y = f(x)$ is a solution to the D.E. with initial condition $f(x_0) = y_0$.

• Its graph passes through (x_0, y_0) .

• $f'(x_0) = F(x_0, y_0)$ by D.E.



- Hence, the slope of the tangent line to $y=f(x)$ at (x_0, y_0) is $F(x_0, y_0)$.

This gives a strategy for sketching the solution curves.

- Plot many points on the plane: $(x_1, y_1), (x_2, y_2) \dots$
- For each plotted point (x_0, y_0) , draw a short line segment with slope $F(x_0, y_0)$.

This method generates a diagram called a slope field, which often gives a general shape of the solution curves.

$$\Rightarrow y(x) = ke^x$$

e.g. A slope field of : (a) $\frac{dy}{dx} = y$; (b) $\frac{dy}{dx} = y - x$;

$$(c) \frac{dy}{dx} = y - x^2$$

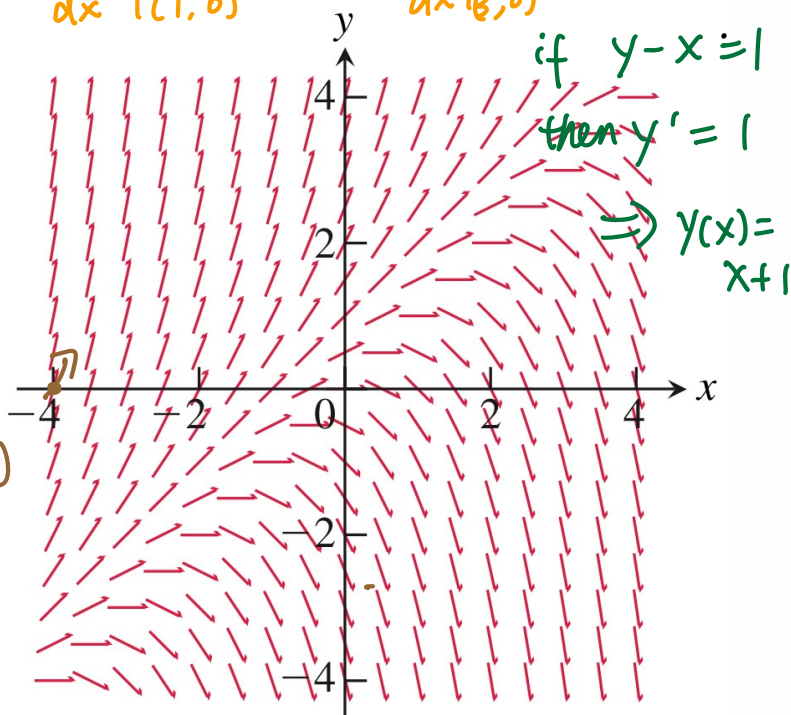
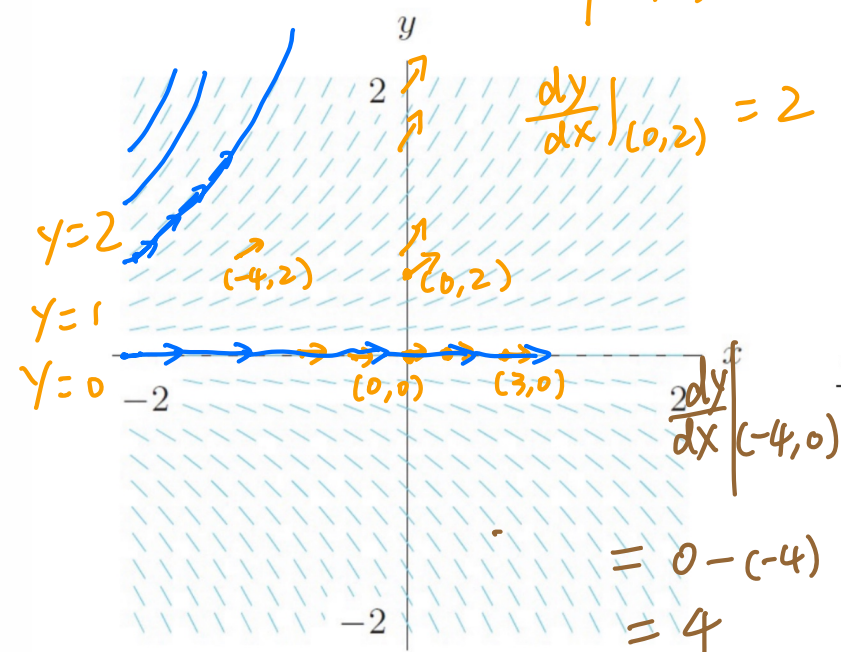
$$\left. \frac{dy}{dx} \right|_{(0,0)} = 0, \quad \left. \frac{dy}{dx} \right|_{(1,0)} = 0, \quad \left. \frac{dy}{dx} \right|_{(2,0)} = 0$$

$$\left. \frac{dy}{dx} \right|_{(0,2)} = 2$$

$$\text{if } y - x = 1$$

$$\text{then } y' = 1$$

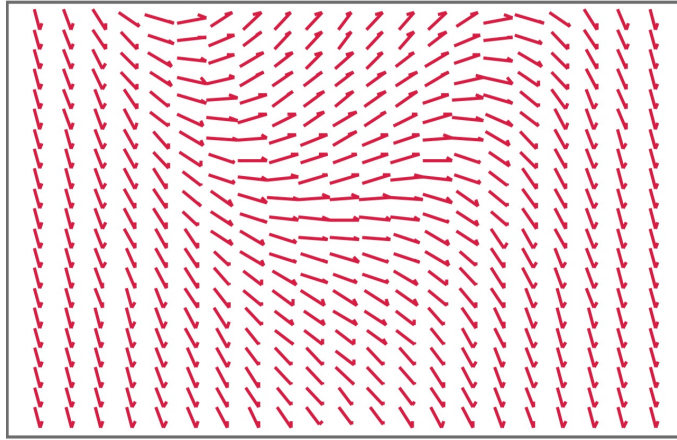
$$\Rightarrow y(x) = x + 1$$



(a) For the same y , the slopes are the same. (b) $y = x + 1$ is one solution

$$y(x) = C \cdot e^x \text{ (later will see)}$$

$$y(x) = 1 + x + ce^x \text{ (later)}$$



$$y=1, \quad x=100$$

$$\frac{dy}{dx} < 0$$

(c) $y(x) = Ce^x + (x^2 + 2x + 2) \quad (\text{later})$